

1 **“Machine to Paint Sounds”**

2 A Creative Music System Based on Collaborative Work Between Teachers and
3 Students

4

5

6 **Abstract**

7 "Machine to Paint Sounds" was an interactive artwork and system collaboratively
8 created by students, teachers, and researchers from the Music programs (Bachelor of
9 Music and Technology, Bachelor of Composition with Electroacoustic Means) at the
10 School of Arts, National University of Quilmes. The project integrated music
11 creation, animation generation, real-time audio processing, musical performance,
12 wireless computer connections, depth cameras, and custom software development.
13 This work was conceived as a tribute to the great musician and inventor Fernando von
14 Reichenbach and his machine, the "Catalina or Graphic Analog Converter."

15

16 **1-Background**

17 "Machine to Paint Sounds" (hereafter MtPS) emerged from research on
18 various technologies related to "touchless" user control interfaces and the real-time
19 processing and generation of sound. As such, it can be considered a musical
20 instrument operated without direct physical contact.

21 The concept of a touchless musical instrument can be traced back to Leon
22 Theremin, who invented the Theremin or ThereminVox between 1918 and 1920
23 (Swingler, 1998a). Leon Theremin's aspirations for his instrument align closely with
24 the goals pursued by MtPS:

25

26 Visualize great possibilities in connection with the problem of controlling
27 sound material by means of other than mechanical, by the free movement
28 of the hands in the air, which offers a more intimate connection of these
29 sounds with the individuality of the performing artist, and will make
30 available not only the expressive power of existing instruments, but
31 perhaps even greater possibilities. (Montague, 1991)

32

33 Furthermore, as discussed throughout this paper, Leon Theremin's concept of
34 using both hands to operate the instrument is directly incorporated into MtPS, serving
35 as a clear reference to the development of the ThereminVox.

36 Leon Theremin, a Military Engineer Radio Officer and a Cellist at the Saint
37 Petersburg Conservatory of Music, uniquely combined these two fields in the creation
38 of one of the most fascinating electronic instruments of the twentieth century. This
39 blend of artistic and technical expertise is also reflected in many of the researchers
40 who contributed to the construction of MtPS, creating yet another parallel with this
41 remarkable inventor and musician.

42 The expressive power envisioned by Leon Theremin is vividly demonstrated
43 in at least two videos, where both the inventor¹ himself and the virtuoso Clara
44 Rockmore² perform on the ThereminVox with remarkable finesse: [Video 1](https://www.youtube.com/watch?v=w5qf9O6c20o), [Video 2](https://www.youtube.com/watch?v=pSzTPG1Na5U&index=2&list=RDw5qf9O6c20o).

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48 ¹<https://www.youtube.com/watch?v=w5qf9O6c20o>

49 ²<https://www.youtube.com/watch?v=pSzTPG1Na5U&index=2&list=RDw5qf9O6c20o>

50 However, the ThereminVox is not the only instrument connected to the work
51 described in this article. The 1980s and 1990s were particularly prolific in the
52 development of various devices that enabled sound generation and processing based
53 on information obtained from hand and body movements, without requiring physical
54 interaction with mechanical or electronic instruments.

55 One notable development is the work of installation artist David Rokeby,
56 specifically his Very Nervous System (VNS, 1982). This system employed a
57 sophisticated computer and camera setup to accurately detect a person's movements
58 and position (Winkler, 1997). The system could interpret both positional and motion
59 data and use them as parameters for sound creation. Remarkably, Rokeby's system
60 achieved an impressive degree of motion detection accuracy, enabling sound
61 generation with even the slightest finger movements. A compelling video showcases
62 Rokeby demonstrating the VNS in action: [Video 3](#).

64 As highlighted in one of the texts featured in the video, Rokeby “...creates a
65 direct, visceral relationship between body, sound, space, and technology.” One of the
66 objectives in developing MtPS was to establish connections between hand movements
67 and sounds similar to those achieved by Rokeby with the VNS.

68 In 1984, Rolf Gehlhaar—a composer, professor of experimental music, and
69 researcher in assistive music technology—developed SOUND=SPACE, a system
70 designed to create an interactive musical environment. Using an ultrasonic
71 echolocation system, it measured the precise positions of people within a space and
72 transmitted this information to a computer that controlled sound synthesizers,
73 triggering sounds and manipulating complex musical parameters in real time
74 (Gehlhaar, 1991).

75 It is worth noting that Gehlhaar collaborated with Luciano Berio and Pierre
76 Boulez in the late 1970s and early 1980s, a connection that likely influenced his work
77 combining music, dance, and technology (Gehlhaar, 1991). The collaboration
78 between musicians, composers, programmers, and sound designers was fundamental
79 in the development of MtPS, emphasizing discussions on achieving the best possible
80 technical, sonic, and aesthetic outcomes.

A video³ demonstrates the functionality of the SOUND=SPACE system:

[Video 4](#)

In the early 1990s, the innovations of Don Buchla—a pioneer in the field of sound synthesizers—stand out, particularly his creation of instruments like the Buchla Lightning (Rich, 1996). The fundamental concept behind these developments was the mapping of MIDI events to hand movements using two wands equipped with infrared light. Joel Chadabe, a composer, author, and internationally recognized pioneer in interactive music systems, described the operation of the Buchla Lightning as follows: two infrared wands emit and transmit information about location, acceleration, velocity, and direction to the photo-sensors of the base unit. The device then translates these actions into corresponding, pre-programmed system data (Chadabe, 1997, p. 231).

Although, in this particular case, an element (the infrared wand) is used to capture hand movement data, what Buchla achieved with such a simple tool is remarkable. For instance, there is a video⁴ featuring percussionist Joel Davel—who used the Buchla Lightning in some of his works—performing a demonstration: [Video 5](#).

In 1998, the most advanced version of the Soundbeam system was introduced. Originally designed by Edward Williams for producing avant-garde dance music, its first prototype was developed in 1984 (Swingler, 1998b). Soundbeam functions as an invisible, elastic keyboard in space, enabling sound creation without the need for physical contact with any equipment. Notably, the project was designed to allow even students with severe physical disabilities to produce auditory and musical effects. Regarding its technical operation, Tim Swingler, director of the Soundbeam Project, explains:

Soundbeam works by emitting an invisible beam of high frequency sound inaudible to human ears. Even very slight interruptions and movements within the beam change digital data about the beam's speed and direction. These data can be interpreted by a wide range of electronic musical instruments. (Swingler, 1998a)

³<https://www.youtube.com/watch?v=tJyutKQEpLk>

⁴<https://www.youtube.com/watch?v=-2juR1usQUQ>

Soundbeam development was inspired by the ThereminVox described above, but as special differences underlie: it does not have a fixed sensing zone and a unique sound does not occur, but it can be adapted to the needs of who uses it (Swingler, 1998a). The official website of Soundbeam⁵ offers an excellent video of the expressive possibilities that the system gives to people with disabilities.

Between 2000 and 2010, three projects shared common features with MtPS: Audio Space by Theo Watson (2005), CaDaReMi by Gehlhaar, Rodriguez, and Girao (2008), and Ocupación Invisible (Invisible Occupation) by the Buenos Aires Sonora group (2010).

The first, Audio Space, is an interactive artwork-system that allows users to navigate a virtual sound space using a head tracker, a pair of headphones, and a microphone. This system measures the spectator's position in real time and uses the data to spatialize binaural sound. A notable feature of this system is that it uses the user's head movement as a reference point, spatializing sound to create a unique auditory environment tailored to each individual—a paradigm distinct from the approaches described in the previous projects.

The Audio Space project page includes a first-person demo of the work along with a technical explanation. Additionally, it is worth noting that Theo Watson is one of the co-creators of openFrameworks (<http://openframeworks.cc>), a collection of C++ libraries utilized in the MtPS project.

Audio Space relates to MtPS in two key aspects: a) the need to generate an audible and visual virtual space; and b) the decision to design the device for operation by a single user.

In 2006, Rolf Gehlhaar led a team including Paulo Maria Rodriguez and Luis Miguel Girao to create the CaDaReMi system, a multi-user interactive sound installation based on Gehlhaar's earlier work, SOUND=SPACE. CaDaReMi addresses a significant limitation of SOUND=SPACE: the lack of visual feedback for users. This absence of visual information makes it challenging for users to establish a spatial reference for their actions. Consequently, SOUND=SPACE requires highly precise auditory skills and significant focus to discern the location of sounds, gestures, and effects. The evolution of SOUND=SPACE into CaDaReMi introduces visual

⁵<http://www.soundbeam.co.uk/>

feedback, enabling users to better understand and interact with the tasks being performed (Gehlhaar et al., 2008).

In this context, MtPS incorporates aspects of the operational logic of CaDaReMi, as it generates and processes sounds based on the same gestures people use to draw on a virtual canvas, as we will explore further in this article.

Finally, *Ocupación Invisible* (Invisible Occupation), an interactive sound installation based on pre-recorded audio from "illegal" radio stations in Buenos Aires, was presented in 2010 as part of the traveling exhibition "Ciudades Ocasiones - Post-it City." This installation utilized a sensing system based on motion sensors (PIR, Passive Infrared), arranged in a grid on the ceiling. These sensors captured data about the spectators' movements, which was then used to play and process audio.

A particularly intriguing feature of *Ocupación Invisible* is the concept that each viewer's body could act as a radio dial, with the potential for interference caused by the proximity of other bodies (Calcagno, 2012). Its connection to MtPS lies in its ability to process audio based on human movement and its use of similar software, such as the Pure Data program. Moreover, this work marks one of the earliest projects by the author of this paper, highlighting an enduring interest in interactive sound installations.

Although this is not an exhaustive overview of all the work done in the development of touchless interfaces for sound generation and processing, the reference works and systems presented in this section provide a foundation for understanding how MtPS was conceived, its unique identity, and the resources it employs. These references highlight a highly creative and diverse field, characterized by a wide range of technologies and the varied needs they address.

The following sections will explore the features of MtPS in detail.

2- Introduction to "Machine to Paint Sound"

MtPS is, at first, a system (instrument, machine) for the creation of music and images in real time, based on three fundamental parts:

1. An animations generator, which provides:

- A fluid simulator.
- A creator of 2D structures, which are affected by gravity and environment.

- A particle system generator.
 - A virtual instrument with individual sounding plates.
2. A real time sound processing system, utilizing pre-recorded sounds and signals captured by microphones. This system allows one to utilize and control effects including a Phaser, a Spectral Delay, a Tremolo and a Granulator.
 3. A motion capture system whose work is to capture the hand movements of a person (in a determined zone of space), turning these movements in controllable variables, and which in turn is able to recognize gestures within these movements.

The architecture of this system enables the convergence and unification of all motion variables extracted from the hands of a director-performer. As a result, a gesture that creates a visual texture on the digital canvas, when processed within the system, also generates changes in sound, driven by the movement parameters used in the sound processors.

MtPS is also, by nature, a collaborative artwork. This concept is supported by the definitions proposed in the theories of Smith (2012) and Kester (2011). According to Smith (2012):

Collaborative methods of practice are increasingly the norm in contemporary art. Such works prioritize processes over object production and technical proficiency, as well as social engagement and community over artistic autonomy. (Smith, 2012,p. 35)

Kester (2011) also explored how collaborative art functions by focusing on ideas of labor and knowledge. He argues that new collective projects produce a distinct form of knowledge—experiential knowledge—that is contingent on the unique exchanges among participants and cannot occur within a scripted event or predetermined object. In a previous article, Kester (2010) pointed out that one aspect of collaborative practice involves a deeper rejection of the "ego imperialism" of artistic identity, achieved through the artist's long-term involvement in a specific site or community.

Based on these definitions, it can be said that it is the collaboration between different holders of knowledge that makes collaborative art thinking a process of

working, creating, and learning. In this context, MtPS stands as a case study of an interesting experience to analyze: the creation of a system for generating music and images that involves varying degrees of expertise and experience between students and teachers. The tensions and releases between the learners and the instructors create an ongoing loop of productive feedback into the work.

It is important to note that MtPS was not created in isolation. It has a precursor: a group called Buenos Aires Sonora, based at Quilmes National University (UNQ). This group was tasked with establishing links between researchers, teachers, and students working collaboratively on more than 10 projects, ranging from sound installation and audiovisual performance to site-specific art (Calcagno, 2013). Beyond this particular case, UNQ research projects have always embraced collaborative work (Calcagno, 2012), offering highly enriching experiences.

From this tradition, in 2015, MtPS was presented as a collaborative and interactive artwork with multimedia components, including image, sound, and performance. MtPS was developed by a team of 24 artists, researchers, teachers, and students who contributed their expertise in computer programming, sound design, sound editing, and the performance of instruments using unconventional techniques.

Teachers and Researchers: Esteban Calcagno (Programmer and General Director), Diego Campos (Sound Designer), Mart ́n Matus (Sound Designer and Composer), Edgardo Palotta (Musician, Sax), Mart ́n Liut (Composer), Diego Romero Mascar ́o (Composer).

Students: Sergio Santoni (Programmer), Fabian Sguiglia (Programmer), Leandro Yabkowsky (Programmer), Sandra Gonzalez (Composer), Ana Gonzalez Gamboa (Musician, Cello), Santiago Vallejo (Musician, Melodica), Ivonneth Rosero (Musician, Voice), Carlos Cattaneo (Sound Designer), Nicolas DelliGatti (Sound Designer), Emilia Monin (Sound Designer, Composer), Carlos Herbas (Sound Designer), Ana Clara Bulla (Sound Designer, Composer), Sol Gomez (Sound Designer), Ignacio Mazzuco (Sound Designer), Luciana Rizzo (Musician, Percussion), Fernando Mastrasso (Sound Designer, Composer), Ariel Cardozo (Composer), Juan Ramos (Composer).

253 This work was prepared for presentation at the University School of Arts,
254 UNQ, during the FASE 7.0⁶ Meeting. FASE is a prominent gathering in Buenos
255 Aires focused on the intersection of art, science, and technology. It serves as a
256 platform for thought, exhibition, reflection, critique, and promotion of the visual
257 arts. The event brings together public and private institutions, both national and
258 international, to showcase research and development in new media, including
259 photography, audiovisual art, electronic art, digital research, animation, internet
260 art, video games, robotics, video art, and biotechnology.

261 In 2015 the theme of FASE 7.0 was the “Work in Progress”. MtPS was
262 presented as an almost completed work, which was used at the time to make different
263 changes and tests, such as the best location in space, noise levels, the audio tracks to
264 be used and some programming issues both visual and sonorous.

265 In addition, this work was presented also in 2015, in the Escalatrónica
266 Festival. Escalatrónica⁷ 2015 was the first non-competitive international electronic art
267 festival to be held in the suburbs of Buenos Aires by Lanus National University. It is
268 known as a gathering venue for artists both local and foreign, students and
269 businesspeople related to the theme proposed. There were conferences, exhibitions,
270 concerts, workshops and development of artworks and products demonstration. MtPS
271 was presented there as a more developed system thanks to the experience in FASE
272 7.0.

274 These meetings provided the foundation for a collaborative musical
275 experiment in which researchers, teachers, and students from the Music and
276 Technology and Composition with Electroacoustic Means programs worked for a
277 month to create the Machine.

278 As a result of this effort, the system developed was capable of creating
279 interrelated music and images, generated to be controlled through the hands of a
280 spectator-performer-director. The system, in turn, could detect specific gestures that
281 allowed for changes in the type of image being generated, while simultaneously
282 altering the sonic textures.

286 ⁶<http://www.centroculturalrecoleta.org/nuevositio/fase-7/>

287 ⁷<http://festivalescalatronica.blogspot.com.ar/>

The gesture detector, as will be discussed later in this paper, was made possible through the use of algorithms developed by researchers at the University of Washington⁸. Figure 1 presents a complete list of the gestures that can be captured by the MtPS system.

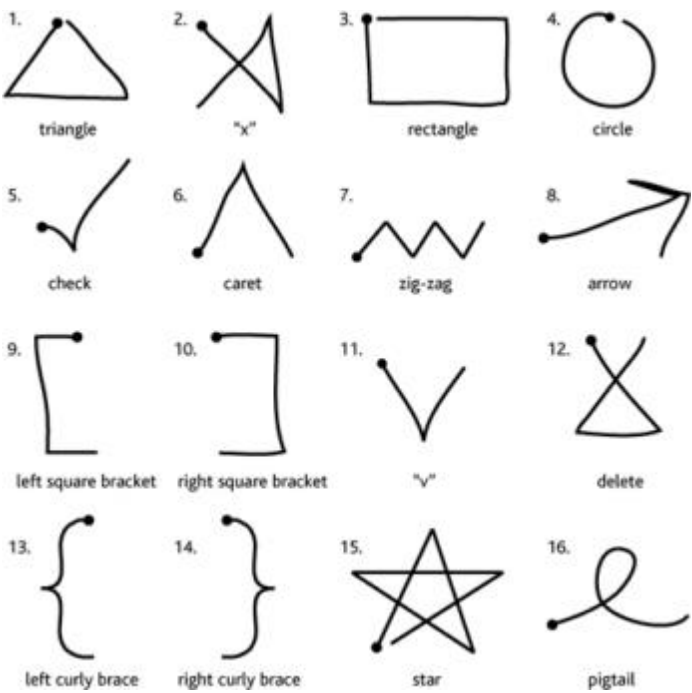


Figure 1: Hand gestures that can be recognized by MtPS. Image taken from <https://depts.washington.edu/aimgroup/proj/dollar/>

For all these reasons, MtPS is a hybrid between a system and an artwork, that is, an artwork-system.

It is important to highlight that this artwork-system was created in honor of the Argentine musician and inventor Fernando von Reichenbach (1931-2005), a pioneer in the combination of music and technology and a teacher at UNQ. His invention, "Catalina or Graphic Analog Converter," capable of generating music by identifying graphical patterns from the analysis of an analog video signal, served as a key inspiration for this work (Del Farra, 2004, p.6). The connection between eternity, creation, invention, and the multiplicity embodied by this artist defined the symbolic direction of the artwork.

⁸<https://depts.washington.edu/aimgroup/proj/dollar/>

The set of elements described in this introduction represents the foundation of MtPS. After providing a general overview, we now proceed with a detailed description of each component of the MtPS artwork-system, including the various technical aspects that shaped the work and the different tools used in each part of it.

3- General Design and Architecture of the Artwork

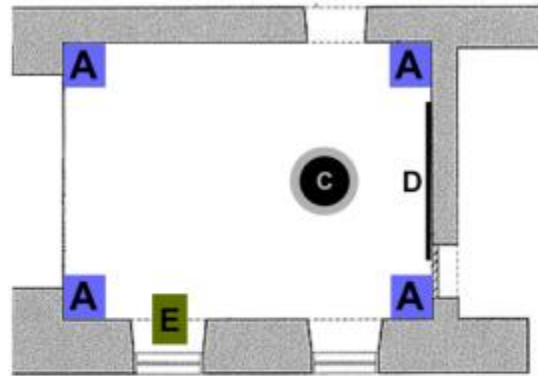
MtPS occupies a 5x5 meter space, as shown in Figure 1.

Figure 2 illustrates the two modes of the artwork: INTERACTIVE INSTALLATION MODE and MUSICAL PERFORMANCE MODE.

In the INTERACTIVE INSTALLATION MODE (Figure 2, above), a circulation space was required, equipped with a fully surrounding sound system, where spectators are placed (one at a time) in front of the machine (C) to interact with it by moving their hands. These movements generated different visual textures, which were simultaneously used to create and process sounds. This mode was primarily used during its presentation at the FASE 7.0 meeting.

In the MUSICAL PERFORMANCE MODE (Figure 2, below), the public and musicians are surrounded by the audio system. Two micro-assemblies are positioned on both sides of the conductor, who is placed at the center of the screen to "paint with his hands" and also to process the audio generated by the instrumentalists. This mode resembled a concert, but with the distinction that in this case, the conductor manipulated the Machine and interacted with the sound produced by the musicians (based on a scheduled improvisation). This mode was employed during the 15 concerts held during the FASE 7.0 meeting.

**Spatial Arrangement in Machine For Painting Sound
Interactive Installation MODE**



**Spatial Arrangement in Machine For Painting Sound
Music Performance MODE**

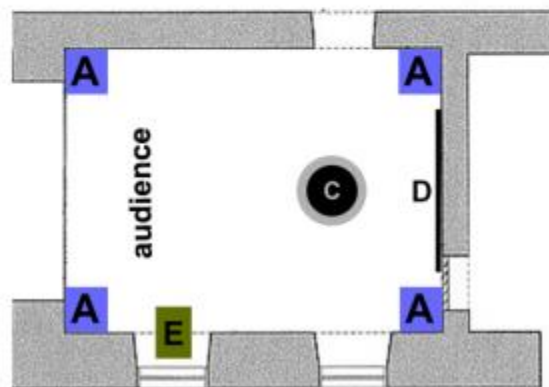


Figure 2: Space in “Machine to Paint Sounds”: above, INTERACTIVE INSTALLATION MODE; below, MUSICAL PERFORMANCE MODE.

4- Aesthetic and organizational development of the artwork

At the beginning of the development of MtPS, an outline for its design and implementation was proposed, featuring key words and concepts similar to those shown in Figure 3.

Machine For Painting Sounds

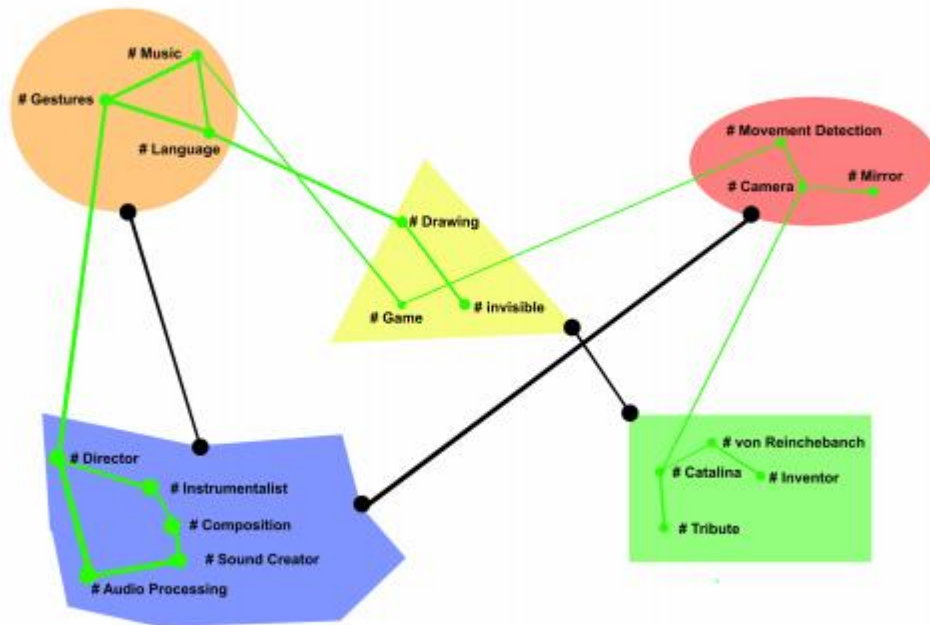


Figure 3: Concepts connection diagram for "Machine to Paint Sounds"

This diagram clearly outlines the core characteristics desired for the proposed system, highlighting key elements related to gesture and movement detection, drawing, music, orchestral conducting, and various concepts inspired by the work of Fernando Von Reichenbach.

Based on this framework, a collaborative generation model was developed, where each group member could contribute their expertise and work collectively with others. Since the project integrated diverse computer technologies to create an interactive experience—such as depth cameras, real-time image and sound generation and processing, wireless device communication, gesture recognition, sound assembly, and sound design—and involved a strong performative aspect, it was essential to divide the system into parts for different groups to work on. This collaborative approach ultimately gave rise to a language-like aesthetic. From this foundation, four action groups were formed:

- 1 **Programming**
- 2 **Sound Material Creators**

3 Sound Design

4 Musical Performance

Each of these groups worked independently, yet collaboratively, to achieve an aesthetic that culminated in the production of a collective artwork.

The Programming group (1) was tasked with researching and developing the necessary computing tools for the artwork. These tools included a detector capable of capturing the spatial coordinates of hand movements in two dimensions, which were then used as data to generate and process both audio and visual elements in real-time. The programming tools utilized by the group were exclusively open-source software.

- openFrameworks⁹, for generating real-time animations, based on different types of Addons that were adapted for the work.
- Pure Data¹⁰, for audio processing in real time from parameters control for a Granulator, a Spectral Delay and a Phaser.
- Processing¹¹, for manipulation of depth camera and software that detect hands and gestures of these hands.

The Sound Material Creators group (2) was responsible for recording the instruments used in the musical performance and selecting the material needed to create music tracks for use in the INTERACTIVE INSTALLATION MODE. This final step was carried out in collaboration with the Programming group (1), as they were also responsible for generating the visual textures in the artwork, aiming to create harmony between the components.

The Sound Design group (3) was tasked with designing the sound system used during the performance. This included decisions regarding microphone types and the choice of playback system, as the entire setup would need to be integrated into the performance space.

Finally, the Musical Performance group (4), consisting of musicians (saxophone, cello, bass clarinet, percussion, melodica, and vocals), was responsible

⁹<http://openframeworks.cc>

¹⁰<http://puredata.info>

¹¹<http://processing.org>

for creating improvisations using extended techniques on their instruments, based on the responses to the generated images and soundtracks.

The four groups began with the premise of working in four stages or phases. Each phase was designed based on words and concepts that could illustrate the personal development of an artist-inventor. These words, which were selected through a purely poetic approach, characterized each of these stages:

- **ETERNITY / ETHER: Eternal creations of an inventor. The place where souls rest before departing the body.**
- **CREATION: The process of developing an invention. Creation as concern and existence of an inventor.**
- **INVENTION: The moment, so to speak, "when the spark ignites".**
- **MULTIPLICITY: The diverse knowledge, skills and qualities typically possessed by inventors.**

An interesting video¹² summarizing what was seen and heard in the INTERACTIVE INSTALLATION MODE can be watched here: [Video 6](https://vimeo.com/151817367).

And another video¹³ displaying one of the concerts in Meeting FASE 7.0 can be watched in: [Video 7](https://vimeo.com/152018793).

5- The system (results)

Technically speaking, the system was composed of several interconnected components, including the following:

- a) A motion and gesture capture system utilizing a Kinect depth camera (Zhang, 2012).
- b) A communication system between computers via OSC and Wi-Fi.
- c) A graphical model generation system: Fluids, Gravity, Particles, and Plates.
- d) A real-time audio processing system.

¹²<https://vimeo.com/151817367>

¹³<https://vimeo.com/152018793>

435
436
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438 Each of these components has unique characteristics, which will be detailed
439 below:

440

441 a) Motion and Gesture Capture System

442 This system utilizes a Kinect depth camera, which is capable of capturing
443 video and measuring the depth of each pixel in the scene. This feature offers
444 significant potential for the project, as it allows for the creation of sensing areas (e.g.,
445 detecting only the pixels within a 2-meter radius from the camera), thereby providing
446 precise control over where motion capture is required.

447 For this artwork, the group employed the Processing programming language
448 (version 2.0) and the SimpleOpenNI library¹⁴ to capture data from the Kinect camera
449 and process the detection of a single person and their skeleton. In this case, only the
450 hand movement data was selected. Additionally, the \$1 Unistroke Recognizer
451 library¹⁵, also available for Processing, was used to detect various gestures made on
452 the digital canvas. As a result, a system capable of detecting individual hand
453 movements and interpreting gestures within a designated space was developed.

454 The complete detection code was written by Fabián Sguiglia (PUNQ I+D
455 "Temporary Space Systems and Synthesis of Sound Art"), a member of the MtPS
456 programming group¹⁶.

457 The capture program was run on Windows 7, as the Kinect depth camera is fully
458 compatible with this operating system.

459

460 To separate the sound reproduction system (including the console, sound card,
461 and video/audio processing computer) from the image projection system (comprising
462 the projector and Kinect), it was necessary to transmit movement and gesture data
463 captured by the Kinect. One of the computers wirelessly transmitted the data to the
464 other. For this purpose, a wireless router (TP Link, 300 Mbps) was used to facilitate
465 communication between the two computers. Once the movement and gesture data
466 were collected, they were sent at high speed via the OSC (Open Sound Control)

467 ¹⁴<https://code.google.com/p/simple-openni/>

468 ¹⁵<https://github.com/nok/onedollar-unistroke-recognizer>

469 ¹⁶<https://github.com/ecalcagno/MachineToPaintingSounds/tree/master/ProcessingCode/gestos2>

protocol. This approach allowed for the separation of the systems, enabling more efficient operation and greater flexibility.

The complete code used for this communication is available on GitHub¹⁷.

c) Generator System of Graphical Models

This system was entirely developed using openFrameworks, a collection of libraries written in C++. In broad terms, four image modelers were created: Fluid, Gravity, Particles, and Plates. In some cases, various types of Add-ons (libraries developed by the openFrameworks community) were utilized and adapted to meet our specific needs.

To develop the fluid simulator, symbolizing the ETHER, we used an add-on called ofxFluid¹⁸, created by programmer-artist Patricio Gonzalez Vivo. This add-on is based on Harris (2012). By developing mathematical fluid dynamics, Harris achieved a highly effective simulation on the GPU of the video card, which Gonzalez Vivo utilized to create a user-friendly add-on for openFrameworks. In the case of MtPS, a 'fluid' is generated from the coordinates of one hand, following its movement, while the other hand determines the color of the fluid. Figure 4 illustrates an example of how the fluid generator operates.

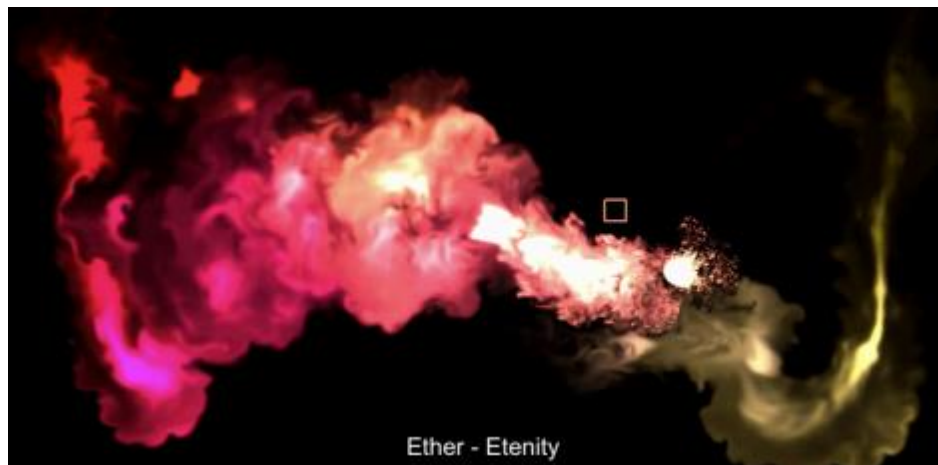


Figure 4: Screen Capture of fluid simulator used in MtPS, representing the concept of ETHERNITY/ETHER.

¹⁷<https://github.com/ecalcagno/MachineToPaintingSounds/tree/master/PureDataCode/communication>

¹⁸<https://github.com/patriciogonzalezvivo/ofxFluid>

For the simulation of "2D rocks" (view Figure 5) which are influenced by gravity and are destroyed when they are rattled against one another or hit on the floor, we used the add-on ofxBox2d¹⁹ a wrapper of Box2D for openFrameworks. Box2D is an open source C++ engine for simulating rigid bodies in 2D, developed by the programmer Erin Catto. This library can emulate an environment where gravity, collision detection, and the displacement of parts of a solid are simulated. In the MtPS, this animation is related to the concept of CREATION²⁰, due to the characteristics of constructiveness that allow the compilation of this library. In this case, the coordinates of a hand are used to generate structures that fall and are broken when they hit the floor. The coordinates of the other hand are used to set up the color of the structures changing the green and red component for RGB color model. This is performed by mapping the position of the hand on the screen (X and Y coordinates) between 0 and 255, which are the values between which the color components are encoded by the program. In Figure 5 we can see a screenshot of how they are created and how these 2D structures behave.

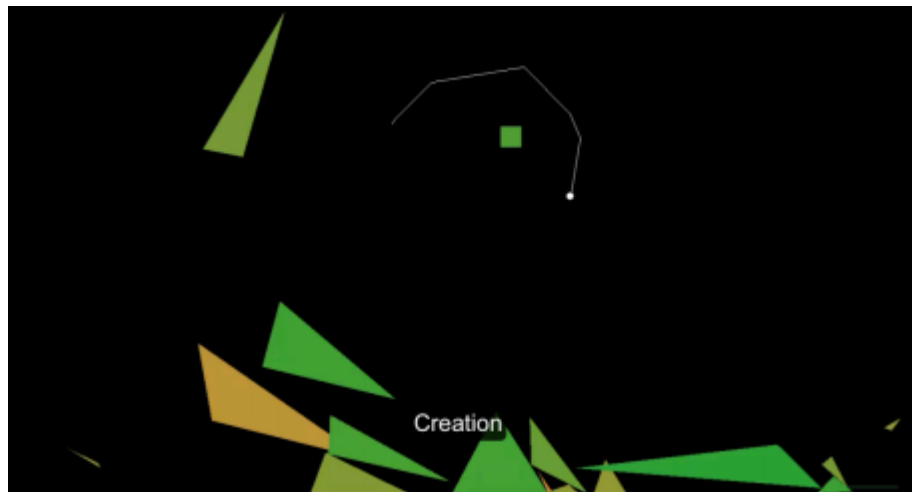


Figure 5: Screen capture of the gravity simulator together with the creator of 2D structures called "2D Rocks", both used in MtPS, representing the concept of CREATION.

The particle system created for the Machine has features that simulate the representation of neural connections. This system is not based on any particular add-on, but it is built from a data structure called a "vector". Thus, hundreds of particles

¹⁹<https://github.com/vanderlin/ofxBox2d>

²⁰Concept explained in section 3 - Aesthetic and organizational development of the artwork.

bind by lines are created. They also move erratically from the center outwards. This creates a sense of continuous movement. This animation is generated from the INVENTION²¹ concept. In Figure 6 an example of how to create the particle generator and how it behaves can be seen.

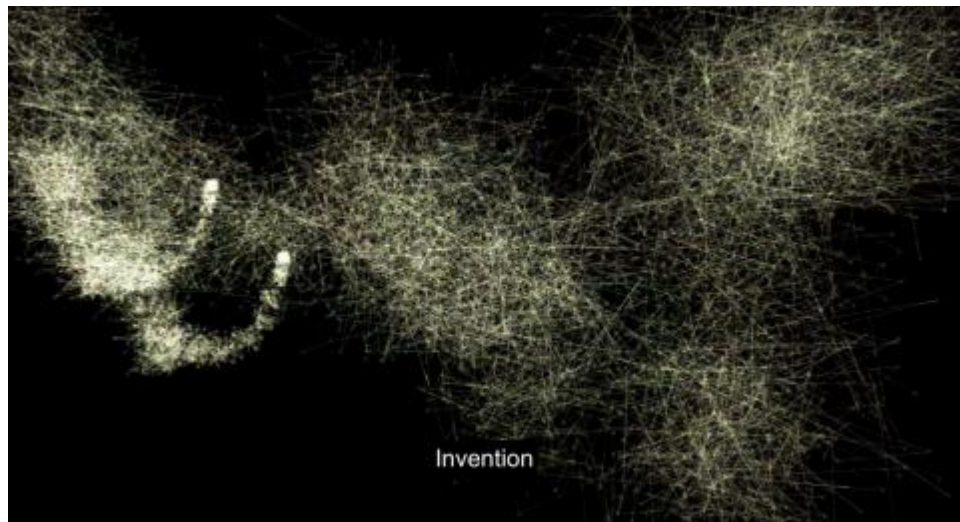


Figure 6: Capture of the particle simulator used in MtPS, which was used for representing the concept of INVENTION.

Finally, representing the concept of MULTIPLICITY²², a virtual plate²³ instrument was created. This instrument was developed using the ofxVoro²⁴ add-on, which can generate a structure and divide it using Voronoi tessellation. The Voronoi tessellation enables the creation of Voronoi diagrams, which subdivide a plane into proximity areas around certain key or special points that define the diagrams. It is a geometric tool widely used in various fields, such as land parceling, antenna-based communication spaces, and more²⁵. In MtPS, this tool is used to divide a 2D polygon into several sub-areas, which are then assigned sounds, allowing the simulation of a bell plate.

²¹Concept explained in section 3 - Aesthetic and organizational development of the artwork.

²²Concept explained in section 3 - Aesthetic and organizational development of the artwork.

²³A thin, flat sheet or strip of metal or other material, typically used to join or strengthen things or forming part of a machine. Instrument Reference: Bell Plate.

²⁴<https://github.com/patriciogonzalezvivo/ofxVoro>

²⁵For more information please consult the article by Kumar and Kurtz (1993).

The ofxVoro add-on, also developed by Patricio Gonzalez Vivo, is based on the Voro ++ library²⁶ which allows the calculation of Voronoi diagrams, which are widely used to make tessellations. The tessellation allows one to completely divide a figure into segments leaving no gaps, or overlappings.

From the implementation of the functions of this add-on, an area with different plates is created, which are identified individually and to which a specific sound recording of some of the instruments used in the performance are assigned. Thus, an instrument of individual plates was created whose sounds when played, respond to the movement of either hand in the area of execution of each plate. In Figure 7 how the different plates are arranged after the tessellation can be seen.

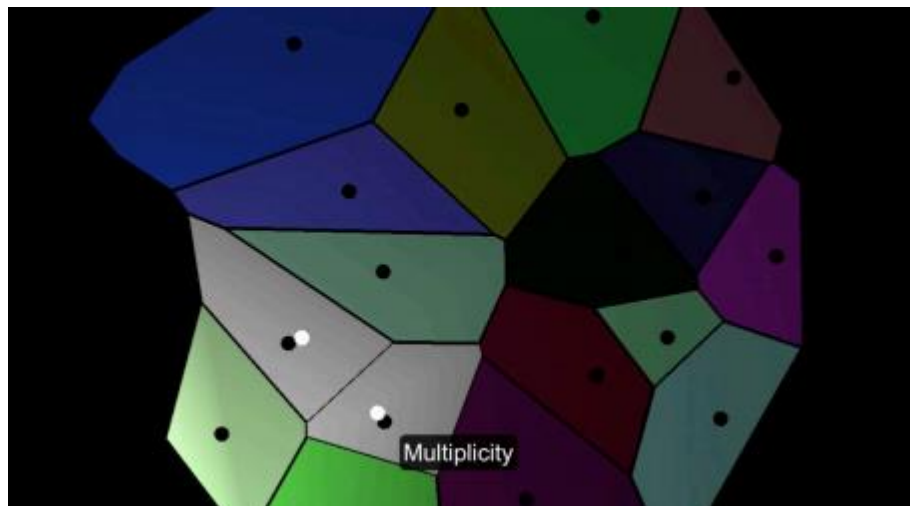


Figure 7: Screen capture of the generator of plates used in MtPS, representing the concept of MULTIPLICITY.

The complete code developed for generating these applications is available on GitHub²⁷.

d) Real-time audio processing system.

This system was developed using the real-time audio processing software, Pure Data. The patch created has several functions: it receives data from the movements and gestures collected by the computer; sends this data to the imaging and animation program; reproduces and processes audio tracks and captured audio from

²⁶<http://math.lbl.gov/voro++/about.html>

²⁷<https://github.com/ecalcagno/MachineToPaintingSounds/tree/master/OpenFrameworkCode>

microphones in real time; and sends the audio signal to be played by the audio playback system.

For collecting and directing the movement variables of the hands, the mrpeach library is used. This library enables sending and receiving data through connections using the OSC protocol. Audio capture and playback are handled using the input and output options of Pure Data (dac~ and adc~), after configuring the sound card to be used. For real-time processing, the following processors were used:

- For the ETHER concept (fluids), a Phaser is used, built from the delwrite~ and vd~ objects in Pure Data. These, together, create a variable delay, modulated by the signal from a phasor~ object, similar to a Low-Frequency Oscillator (LFO). This process generates a slow sound texture with a great variety of internal dynamics, which is considered appropriate to complement the fluid generation. The coordinates of each hand control the speed, depth, and feedback of the LFO.
- For the CREATION concept (structures affected by gravity), a granulator (algorithm that uses granular synthesis techniques for generating an audible sound texture) was used, specifically an object bubbler+ of the SoundHack²⁸ library for Pure Data. The granulator can generate microsounds from the audio track, which have different sizes (in terms of size of time sample) and pitch, which may be considered relevant to complement the animation generated. The coordinates of both hands allow one to modulate the reading time of each grain (each of the sound samples generated by the granulator are called grains), grain density (number of grains per time unit) and grain size (grain size in time)²⁹.
- For the INVENTION concept (the noisy particle system), a spectral delay was used. This was constructed using Pure Data objects to perform Fourier analysis, specifically rfft~ and irfft~, in conjunction with previously used objects: delwrite~ and vd~. In this configuration, the coordinates of each hand affected the delay of each frequency band, resulting in the

²⁸<https://puredata.info/downloads/soundhack>

²⁹For a broader description of the features and operation of a sound granulator you can review the text of Roads (1998).

displacement of the bands and the reconstruction of the prerecorded sounds with every movement.

The complete code used to generate these applications is available on GitHub.³⁰.

Through the creation of all these tools and their subsequent integration, we were able to build the system that powered MtPS.

6- Conclusions

"Machine To Paint Sounds" emerged from the collaborative work of students, faculty, and researchers from two academic programs: the Bachelor of Music and Technology, and the Bachelor in Composition with Electroacoustic Means, both from the School of Arts. It was also the result of three research and development projects: "Digital Developments Applied to Art", "Contemporary Music Territories in Argentina (1973-2010)", and "Temporary Space Systems and Synthesis of Sound Art" at Quilmes National University. Without the involvement of these contributors, it would not have been possible to integrate all the technical and conceptual elements required for the artwork's realization.

In this context, the artwork-system invites us to view music not only as a sound phenomenon, but as a collective creation. While computing technologies allowed us to overcome various technical challenges in developing the system, it was the social interactions that enabled the convergence of different perspectives on music creation, merging them into a unified artistic outcome.

What is particularly interesting about MtPS is that it does not limit itself to the creation of a single musical piece. Instead, with every spectator's interaction and every scheduled concert, a new artwork is born, each one connected to the creative contributions of all the artists involved.

³⁰<https://github.com/ecalcagno/MachineToPaintingSounds/tree/master/PureDataCode>

Thus, the concept of this artwork-system fosters the fusion of collective musical creation with computer technology. It not only generates a work of art, but its very existence relies on and reflects the musical ideas of each participant, making the entire process an ongoing collaborative effort.

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